



The Problem & Approach

Why This Question?

Data curation is increasingly central to multimodal question answering and reasoning, but its effects are hard to isolate. Many reported gains combine changes in model architecture, training recipe, supervision style, and data composition, making it unclear which part of the pipeline explains the improvement.

We study a narrower question: with the model and fine-tuning protocol fixed, which data-curation choices matter most? The NeurIPS 2025 DCVLR challenge is our testbed — it fixes the base model and training procedure and varies only the curated training set, isolating data-efficient multimodal reasoning under realistic resource constraints.

Our goal is not a new architecture or a universal data-selection algorithm, but a controlled empirical characterization of which curation signals work in this fixed-protocol regime.

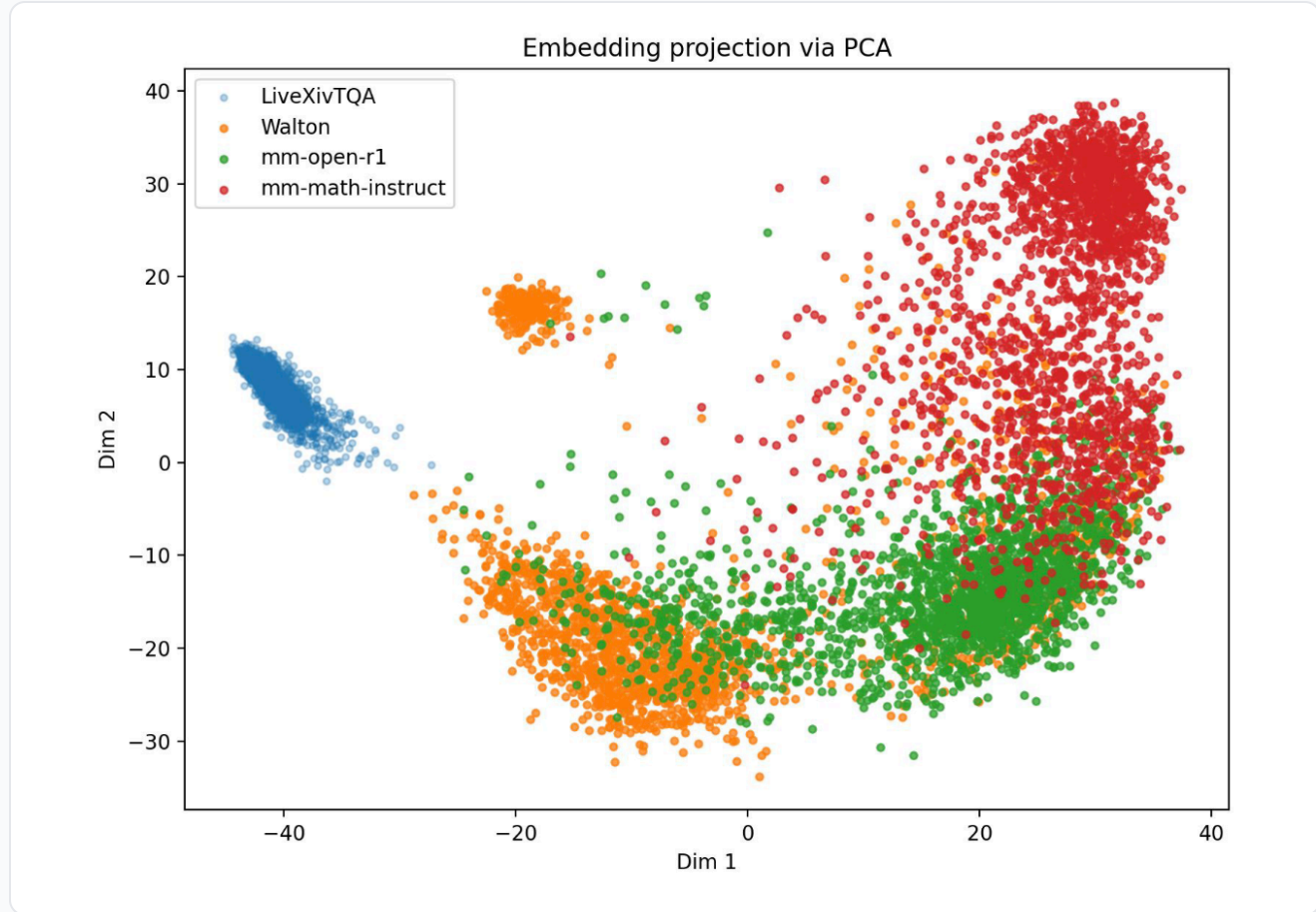
Our 1k-example submission improved the DCVLR aggregate

38.4 → 46.1

+7.7 points from data curation alone
1,000 examples from Walton Multimodal Cold Start

Pick an Aligned Source

Before filtering, the source pool must be aligned with the benchmark. In Qwen2.5-VL-7B embedding space, Walton Multimodal Cold Start clusters near LiveXivTQA — the dominant benchmark. MM-Open-R1 and MM-MathInstruct do not.



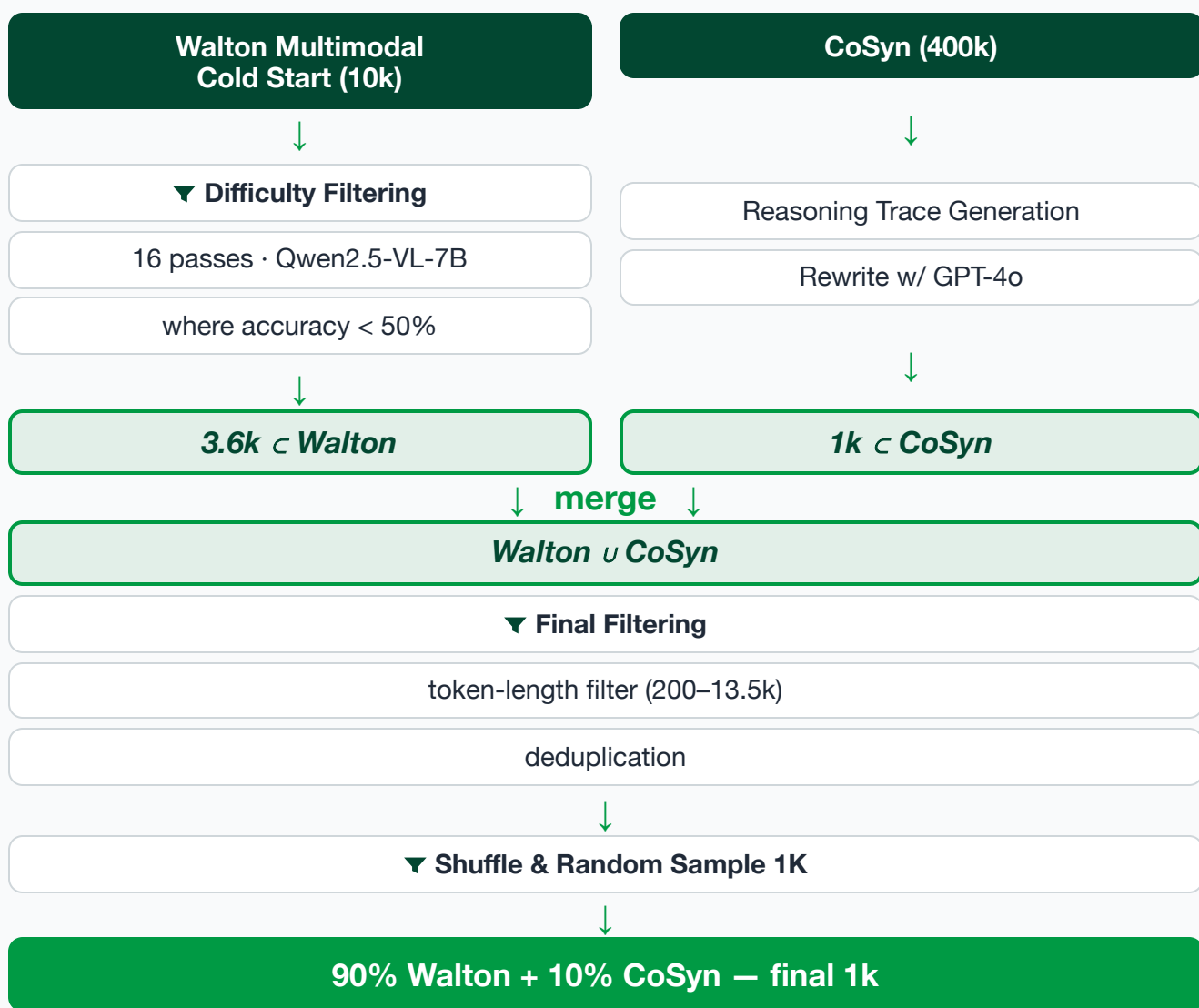
PCA projection of Qwen2.5-VL-7B embeddings (Fig. 1). Walton lies closest to the LiveXivTQA cluster, motivating it as the starting corpus.

Score by Model-Relative Difficulty

For each Walton example, run Qwen2.5-VL-7B-Instruct for 16 rollouts (T=0.7, top-p=0.9). Let k = correct rollouts out of 16:

$k \geq 15$ — Easy	already mastered; little signal
$k \leq 3$ — Hard	unstable or off-distribution
$k \leq 8$ — Moderate	challenging but learnable ✓

Our Submission



Results & Evidence

Difficulty Filtering Beats Every Alternative

At fixed 1k examples, **moderate difficulty ($k \leq 8$) wins**. Too easy or too hard both hurt.

Subset	Threshold	Mean ± Std
Hard only	$k \leq 3$	0.472 ± 0.023
Moderate	$k \leq 8$	0.491 ± 0.002
Easy only	$k \geq 15$	0.440 ± 0.024
Random Walton	—	0.475 ± 0.010

Per-Benchmark Confirmation

Benchmark	Base	Random 1k	$k \leq 8$
LiveXivTQA	0.568	0.779 ± 0.011	0.788 ± 0.002
OlympiadBench	0.134	0.112 ± 0.009	0.144 ± 0.002
Physics	0.095	0.091 ± 0.016	0.098 ± 0.007
VMCBench-DEV	0.798	0.787 ± 0.006	0.790 ± 0.003
Omni3DBench	0.348	0.341 ± 0.005	0.341 ± 0.003
Overall	0.384	0.475 ± 0.010	0.491 ± 0.002

Full Competition Results

Benchmark	Samples	Base	Ours 1k
LiveXivTQA	7,913	56.8	74.5
OlympiadBench	5,929	13.4	11.7
VMCBench-DEV	1,000	79.8	78.2
Omni3DBench	501	34.8	32.6
Atomic	200	8.5	11.5
Statistics	240	17.1	14.2
Overall (weighted)	16,640	38.4	46.1

Large gain on the aligned task (LiveXivTQA); small mixed changes on held-out tasks reflect a specialization–generalization trade-off.

Does the Signal Transfer Across Models?

The same Qwen-derived difficulty scores, applied to train three other architectures (accuracy %):

Target model	Random 1k	Diff.-filtered	Δ pts
Molmo-7B	18.2	23.3	+5.1
Phi-4	31.4	32.7	+1.3
Gemma3-4B	31.6	30.7	−0.9

More Data Stabilizes; It Doesn't Lift the Mean

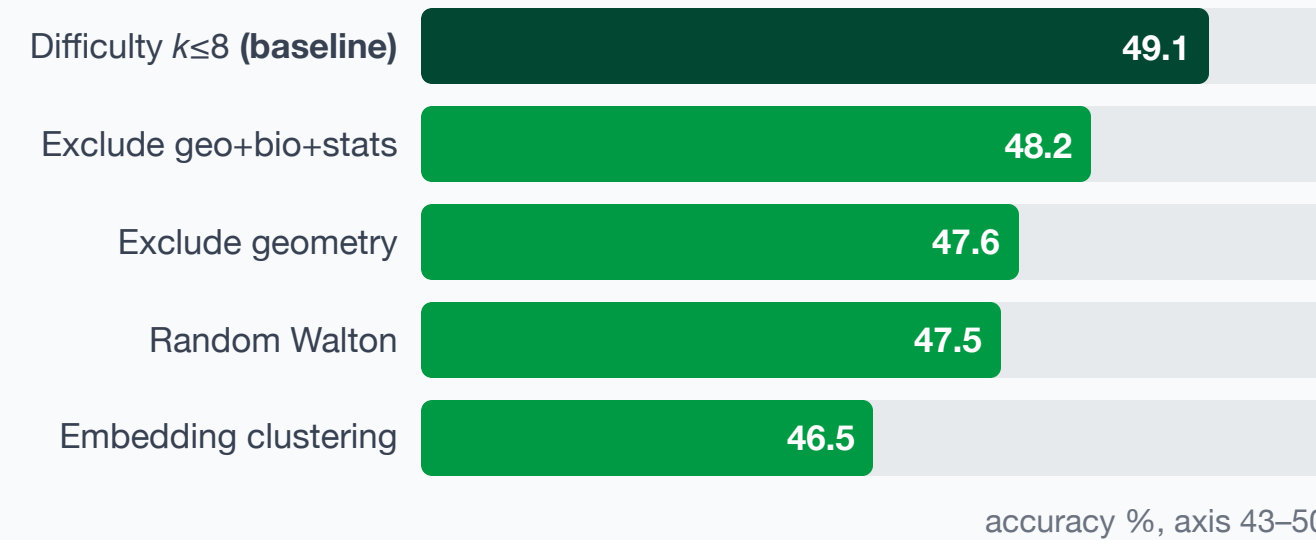


	250	1k	5k	7.5k	10k
Accuracy	.470±.020	.475±.008	.472±.004	.469±.003	.467

Negative Results & Recipe

What Didn't Help: Diversity Heuristics

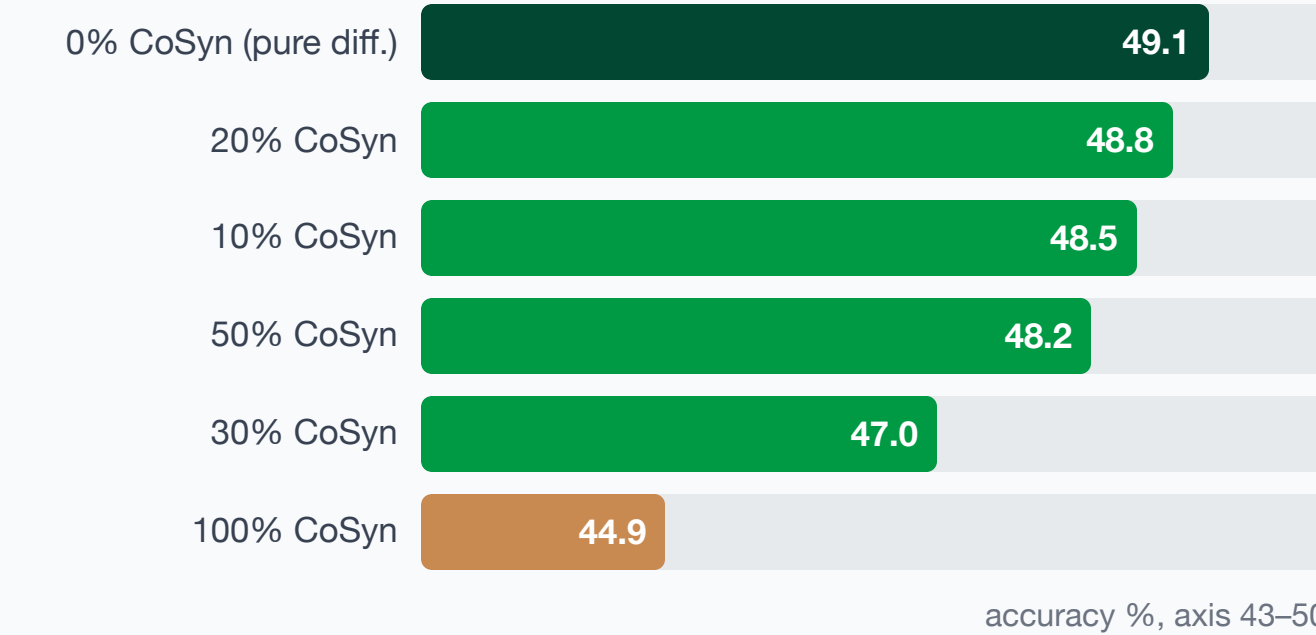
On Walton, none of nine diversity variants (embedding clustering, category balancing, topic exclusion) beat the difficulty-filtered baseline. Some beat random; none reach it.



Replicated on a second source (**Open-R1**): difficulty-filtered **46.4** still leads embedding clustering (45.3), topic exclusion (45.3), and difficulty + topic exclusion (44.4).

What Didn't Help: Synthetic Rewriting

Mixing GPT-4o CoSyn reasoning traces into the difficulty-filtered Walton set: the more synthetic data, the worse it gets.

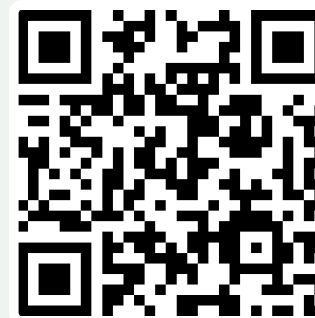


These are negative results about these specific implementations — not claims that diversity or synthetic data can never help.

A Four-Step Recipe

- 1 Choose an aligned source.** Measure embedding proximity between your pool and the benchmark. Alignment unlocks all downstream gains.
- 2 Filter by model-relative difficulty.** Select the challenging-but-learnable zone ($k \leq 8$ of 16 rollouts). Avoid the trivial and the unlearnable.
- 3 Treat scale as a stability knob.** Past ~1k examples, more data reduces variance, not mean error.
- 4 Ablate diversity & synthetic augmentation.** Test at matched scale; additive benefit is not guaranteed.

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